Learning Deep Resonant Prior for Hyperspectral Image Super-Resolution

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***Abstract*— The hyperspectral image super-resolution (HSISR) task has been widely studied, and significant progress has been made by leveraging the deep convolution neural network (CNN) techniques. Nevertheless, the scarcity of training images hinders the research progress of the HSISR task. Moreover, the differences in imaging conditions and the number of spectral bands among different datasets make it very difficult to construct a unified deep neural network. In this article, we first present a nontraining-based HSISR method based on deep prior knowledge, which captures the image before restoring the high-resolution image by using the intrinsic characteristics of CNN. Then, we append a special network input processing module (IPM) onto the HSISR network to automatically adjust the structure of the input so that the choice of network structure is no longer limited, while the network design focuses on exploiting the spatial information of hyperspectral images (HSIs) and the correlation between spectral bands, making the method more suitable for HSISR tasks and greatly extending its applications. Extensive experimental results on the HSI datasets illustrate the effectiveness of the proposed method, and we have got comparable results with the state-of-the-art methods while requiring no training samples.**

***Index Terms*— Deep convolutional neural network (DCNN), hyperspectral remote sensing, image super-resolution.**

I. Introduction

HYPERSPECTRAL images (HSIs), which contain rich spectral and spatial information, have the advantage of capturing the details of the target object. They have a very wide range of applications, such as the observation of the land, atmosphere, ocean, and other fields. However, due to the limitations of incident energy, there is always a compromise between spectral resolution and spatial resolution in the design of optical remote-sensing systems [1]–[3]. Thus, if all other factors stay constant and a high signal-to-noise ratio (SNR) is ensured, the spatial resolution decreases as the spectral features increase. Therefore, HSI super-resolution (HSISR), which aims at improving the spatial resolution of HSIs, is an important part of HSI processing in real-world applications.

Numerous HSISR methods have been investigated for decades, including early classical interpolation-based approaches, fusion-based approaches, and the latest deep learning-based approaches. The interpolation-based method ([4]) directly used the information of neighborhood pixels, which does not bring in much extra information and thus cannot produce satisfactory results. Besides, the Bayesian reasoning and sparse representation-based fusion method fuse the observed low-resolution HSIs with auxiliary images (such as panchromatic, RGB, and multispectral images) to improve the quality of low-resolution images. Although these models can achieve considerable performance, they all require well-registered images, which severely restrict the applications of fusion-based approaches in the real world.

With the rapid development of convolution neural networks (CNNs), deep learning-based methods have been developed vigorously in recent years [5], [6]. Early HSISR approaches improve super-resolution accuracy by applying super-resolution CNN (SRCNN) to HSISR tasks, but these methods often neglect the spectral correlation of HSIs [7], [8]. Recent studies have shown that the application of employing attention mechanisms in HSI recovery tasks has also developed tremendously. Among them, Yao *et al.* [9] exploiting the powerful detail extraction capability of attention mechanisms to establish long-range dependencies between image contents in networks using the self-attention mechanism, and Qu *et al.* [10] using Cross-Attention to produce more efficient spatial–spectral information transfer in the network by adaptively learning spectral and spatial response functions from the HS-MS correspondence. The transformer networks proposed by Bandara and Patel [11] improve the generalized sharpening by learning the spatial dependence and long-range details of cross-features of PAN and LR-HSI spatial and spectral quality metrics of HSI. Preserving the spectral information and considering the correlation between spectral bands are

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essential to avoid spectral distortion [12], [13]. Unlike traditional gray/RGB images, due to the internal characteristic of HSIs, the super-resolution task will inevitably face the following challenges.

1) it is difficult to obtain HSI prior information. HSIs usually contain hundreds of continuous spectrum bands. Therefore, to avoid the over-fitting problem, it is necessary to learn the real prior images from lots of sample images. However, compared with natural images, the number of available HSI datasets is very small.
2) it remains a challenge to construct a unified deep neural network for the HSISR task. Considering the differences in imaging conditions and the number of spectral bands of HSIs, image super-resolution becomes ill-posed due to the one-to-many mapping.

Recently, Ulyanov *et al.* [14] illustrated that the bulk of the image statistics is captured by the structure of the generator CNN, which is independent of the learning process. It means that the learning ability and the internal structure of CNN are both conducive to the processing of image data. Moreover, this also indicates that the structure itself captures and processes image statistics without learning the "meta-details" shared between external images. All these characteristics have attracted many researchers to explore it. Sidorov and Yngve Hardeberg [15] first applied the DIP algorithm to HSI SR, which is an expansion of the field of the task of HSI restoration. Zheng *et al.* [16] used this algorithm as a preprocessing means to upsample LR HSI, preserving their spatial and spectral information and then concatenated upsampled result with the panchromatic image as the input to the network, achieving good performance. Bandara *et al.* [17] further improved the quality of up-sampled LR HSI with this algorithm by introducing an additional spatial domain constraint. Liu *et al.* [18] propose a novel unsupervised framework for fusing the observed LR-HS and HR-RGB images and construct a loss function to learn priors in the latent HR-HS image. Uezato *et al.* [19] proposed a guided deep decoder network as a general prior by embedding multiscale features of guided images into the deep decoder network, which achieves state-of-the-art performance. However, the application of the above methods to DIP is still basically based on its own U-Net structure. In fact, the network structure needs to match the input to produce better results, and the choice of network input is worth exploring deeply.

Nevertheless, applying the deep image prior (DIP) [14] algorithm directly to HSIs can also cause a number of problems: 1) the proposed structure is based on the traditional gray/RGB image super-resolution task, so the method cannot achieve satisfactory results for processing high-dimensional HSIs and 2) through experiments, it shows that the network structure requires a matching input to produce better results. Due to the lack of research on network input and network structure settings in the application of the DIP method in HSISR task, most typical super-resolution networks cannot get satisfactory results, such as ResNet [20], Densenet [21], VDSR [22], and LapSRN [23]. Based on this, we proposed the DRPSR algorithm, which absorbs the advantages of DIP work [14], which means that its application is not limited by the differences in imaging conditions of training data and the number of spectral bands, because we obtain the high-resolution HSIs in the iterative process. In order to solve the shortcomings of DIP applied to the HSISR task, we proposed spectral attention residual module (SARM), which is composed of a spatial residual module and a channel attention (CA) module. The former pay more attention to the high-frequency information and the latter focuses on the spectral correlation, which together forms the SARM to make full use of the spectral correlation and spatial information of hyperspectral data. SARM makes the overall network structure more suitable for processing hyperspectral data. Besides, we put forward an input processing module (IPM) to automatically adjust the input of the proposed network structure. It makes the input noise quickly produce "resonance" with the subsequent network and also encourages the above-mentioned network structures to greatly improve the application performance of the DIP algorithm. Researchers can restore HSIs without the limitation of datasets. The proposal of IPM expands the application of this method and brings the gain of the restored image quality, which breaks the limitation of the DIP algorithm.

We summarize the main contributions of this article as follows.

1) One network architecture is designed for the HSISR task, which can make full use of the spatial information and the spectral correlation of the HSIs. Especially, the spectral correlation information is proved to contain useful underlying prior image information.
2) On top of the deep prior knowledge-based method, we innovatively put forward one special network IPM to make the DIP method much more suitable for the HSISR task, which overcomes serious limitations of network architecture selection and then further improves the quality of the high-resolution image, without requiring the training samples.
3) Compared to methods that require training, this study can achieve almost the same performance as the trained method in a relatively short time without the need for ground-truth reference. We propose the DRPSR method to solve the difficulties of unclear reconstructed images and low indicators caused by insufficient training data in small datasets and improve the application of the DIP method.

The rest parts of this article are organized as follows. The related work is introduced in Section II. We elaborate our method in detail in Section III. In Section IV, we conduct comprehensive experiments on CHIKUSEI [24] and PAVIA CENTRE datasets to evaluate the proposed method, which shows that our approach has a better performance than many other methods. Finally, the conclusion of our work is presented in Section V.

II. Related Work

In this section, we briefly review some of the recent advances that are most relevant to our work in the field of image super-resolution. These methods can be grouped into two categories: 1) external information-based SR methods and 2) internal information-based SR methods.

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A. External Information-Based Methods

The methods based on external information restore the high-resolution image by the "meta-detail" shared among external images. The super-resolution methods via deep learning are the classic methods based on external information, which have achieved remarkable performance at present. The CNN-based method for image super-resolution was first proposed by Dong *et al.* [25]. Thereafter, various architectures based on it have been investigated for image super resolution, such as deeply-recursive CNN method [26], [27], very deep super resolution network [22], super-resolution generative adversarial network [28], and Laplacian pyramid super-resolution network [23]. All the above methods simulate the mapping between the low-resolution image and high-resolution image through a lot of training data. Given a large sample pair of the original image and its low-resolution version, you can learn the deep network to restore the original high-resolution image from the degraded one. Nevertheless, most existing SR approaches are designed for monochromatic or color images. Due to neglecting the spectral correlation, applying these methods directly to the HSISR task would often result in spectral distortion.

In recent years, HSISR technology has received much more attention. Since CNN has remarkable abilities to simulate the mapping between LR (low resolution) images and HR (high resolution) images, many CNN-based remote-sensing approaches have flourished recently. Among them, Liebel and Korner [7] improved the SR performance by applying SRCNN [25] to the HSISR task, which is an important milestone. Nevertheless, the differences between HSIs and RGB images were not considered, so the fineness of the reconstructed image is reduced for this reason. Inspired by this, a 3-D full CNN is developed by Mei *et al.* [12] to reconstruct the high-resolution HSI, which not only explored spatial information, but also considered the spectral information. Li *et al.* [13] used CNN to learn spectral differences among the adjacent bands and also proposed the SCT-SDCNN approach, which adopts a spatial constraint strategy to make low-resolution images similar to HR images. Jia *et al.* [29] utilizes spatial context in neighboring areas to enhance spatial resolution and also considered the spectral correlation in neighboring bands. Thereafter, in order to enable the network to better extract the potential spatial and spectral information of the image, Wang *et al.* [30] designed a spectral–spatial residual module that adaptively learns more effective features from all hierarchical features in units through local feature fusion, significantly improving the performance of the algorithm.

However, it still remains challenging to apply CNN-based methods to a real HSISR system. Because HSIs could be acquired from different hyperspectral cameras, and it is not easy to design a unified deep network architecture under different spectral band numbers and imaging conditions.

B. Internal Information-Based Methods

The most intuitive method is based on interpolation by using the information of neighboring pixels directly. The interpolation-based method is one of the fundamental SR methods, which is very simple and easy to implement. However, this method cannot introduce too much extra information and can hardly restore the high-frequency details for the high-resolution images. Rudin *et al.* [31] proposed the total variation (TV) super-resolution reconstruction method, which can effectively remove noise and blur, but in the case of strong noise, the smooth area of the image will produce a step effect, and the texture information of the image cannot be well retained. Farsiu *et al.* [32] proposed the two-side TV (BTV) regularization method. This approach not only considers the geometric distance between the surrounding pixel and the center pixel, but also considers the grayscale similarity, which greatly improves the performance of the BTV algorithm compared with the original TV algorithm. Most of the above methods model the inverse tasks of these image recovery as energy minimization problems and are formulated as follows:

$$x^* = \min_x E(x, x_0) + R(x) \tag{1}$$

where $E(x, x_0)$ denotes a metric related to the task, and x and x_0 are the original image and its corresponding LR image, respectively. $R(x)$ denotes a regularizer, we can learn it from data or just chose it manually. Nevertheless, Ulyanov *et al.* [14] have shown that through the optimization process, priors can be found directly in the network parameter space, which allows for the removal of $R(x)$. The solution can be searched as

$$\theta^* = \operatorname*{argmin}_\theta E(d(f_\theta(z)); x_0), \quad x^* = f_{\theta^*}(z) \tag{2}$$

where $f_\theta(\cdot)$ denotes the CNN model with parameters θ, $d(\cdot)$ stands for degenerate operation, θ^* is the obtained optimal parameter, and z denotes an input of random noise. Therefore, the HR image will be restored by optimizing the network using only the LR image. Ulyanov *et al.* [14] describe that in the learning process, a generator network is structured to learn a large number of low-level image statistic priors and then the network structure will automatically reject the noise and give priority to learn image details. Therefore, the optimization process should be stopped in time when the iteration time reaches the appropriate number. This method also narrows the gap between two image restoration methods: the learning-based method using CNN and the nonlearning-based method utilizing hand-crafted image priors, but it has not been thoroughly studied. The reconstruction task has worked well on the structure of the "hourglass" (also referred to as "decoder–encoder"). Hourglass structure resembles U-net [33] [34], but has less number of skip operators and BatchNorm before the concatenation operator. They got 41.95 PSNR on the benchmark of Plotz and Roth [35]. It is even comparable to some deep neural network methods that have been trained with a lot of training samples, but the reconstructed image still does not achieve a high level of spatial information recovery. The network input takes fixed input z (random noise), which is not a conventional input for the images. It means that the image statistics can be captured by the structure of ConvNets, which validates the importance of the generator network architecture.

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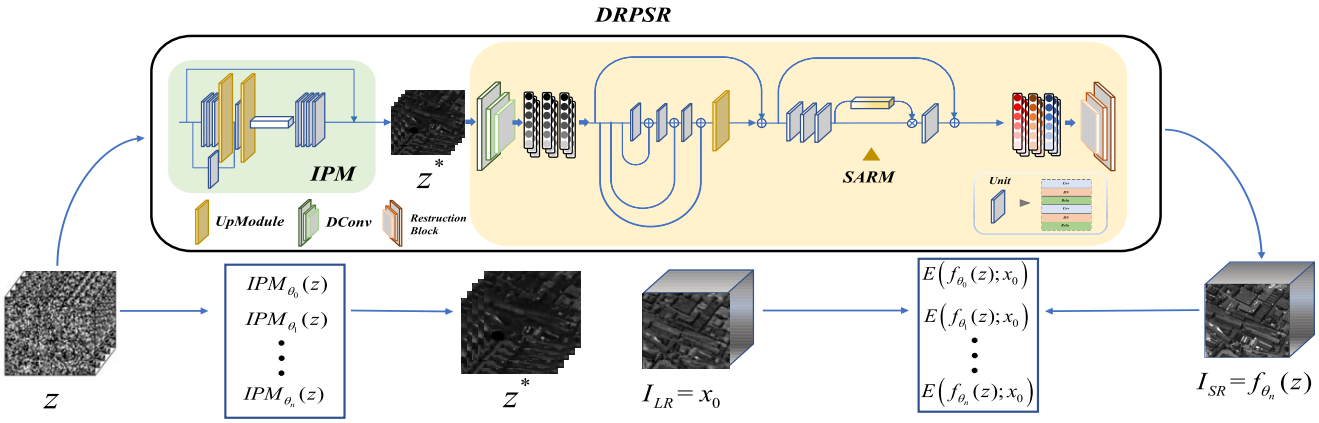


Fig. 1. Network structure of the proposed DRPSR network, where "IPM" and "SARM" denote input processing module and spectral attention residual module, respectively.

This approach has been used on classic network models such as Resnet [20], but the PSNR is only 31.95 dB, but get 35.05 PSNR for UNet, that is to say, the superposition of concatenation operators in the network can improve the performance. In contrast, many effective SR methods are not applicable, such as ResNet [20], DenseNet [21], VDSR [22], and LapSRN [23], the reason of which still needs to be explored. Sidorov and Yngve Hardeberg [15] obtained good performance in the HSI restoration task by using a DIP network structure, and it proved that the 3-D implementation is not suitable for this method, which needs to take much more memory and computational time. However, the network structure is still not fundamentally changed. The connection part is basically the same as the concatenation operator, and a simple change to the network structure often results in a significant performance decline. In view of this, we developed the DRPSR method, whose advantage is that we can choose any network structure to maximize various types of ConvNets. Through extensive experiment results, We have found that the performance of HSISR is obviously dependent on the "resonance" between the network input z and the specific network structure. In the application of the DIP algorithm, if an input can achieve the desired performance of the network after it is put into the network, then the input can be said to "resonate" with the network structure. If a network structure shows good performance in a training-based approach, but performs much less than expected when applied to the DIP algorithm, it may be caused by the mismatch between the input and the network structure. In the above work, there is a less in-depth study on how to adapt the network inputs to the network architecture. In fact, different from the previous methods that take the low-resolution image as the input, the selection of input is also very important for this algorithm because of the uncertainty in input selection. Obviously, the constraints on the selection of convolutional network structures will greatly reduce the quality of the recovered image. Given that it is still impractical to artificially set different input z for different network structures based on experience, we propose a module to automatically adjust the input z. It can automatically adjust the input according to the network built later, so that the input can be compatible with the network. The experiment shows that the addition of residual blocks in the DIP method often reduces the quality of the SR image. In contrast, the hourglass structure is an appropriate network structure for random noise input. We also found this through multiple experiments. However, when we add an IPM to process the input, even the Resnet network structure that has previously restored the image least effectively improved the final result by leaps and bounds, surpassing the image quality produced by the hourglass structure. On the other hand, if this module is added, the image quality of the hourglass structure recovered will be better. In subsequent experiments, we can prove that this module can indeed find the optimal input that resonates with the network structure, so that the choice of network structure is no longer limited.

III. Methodology

A. Network Architecture

In Fig. 1, we present the network structure of the proposed DRPSR method, which mainly consists of three parts: the input processing block, feature extraction block, and reconstruction block. As we mentioned above, among the classical network architectures that apply this approach, the U-net with lots of concatenate operations tends to perform better, whereas the network with residual-dominated structures tends to perform the worst. However, the performance of these structures is severely limited, because the input module of the structure is complicated, and the subsequent networks need to "resonate" with it when applying this iteration-based DIP approach. In view of this, we design the IPM (Fig. 2) to optimize the input Z into a Z^* that "resonates" with subsequent networks, the latter network architecture will no longer be limited.

The first IPM part has three branches. We leverage three Conv layers (k3-c128-s2-p1) to form the main branch to reduce the size of the feature map. We used two upsampling modules to enlarge the feature maps, step by step. After passing through the SARM in IPM, all the branches converge. Before feeding the features to the first unit, we add three Conv layers (k3-c102-s1-p0) to do the initial feature extraction operation.

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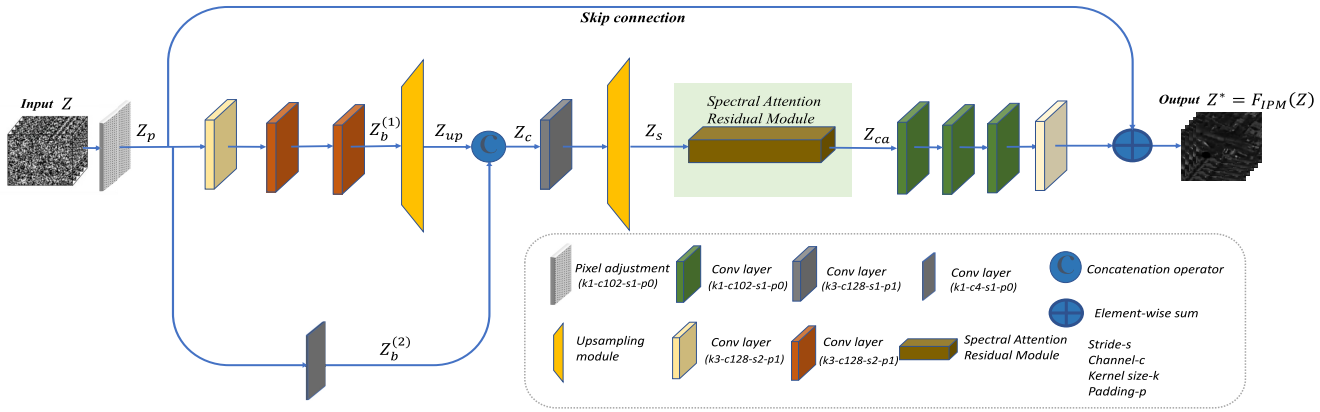


Fig. 2. Network architecture of the IPM, which mainly consists of three parts: "PA," "channel adjustment," and "SRA."

The recently proposed residual network structure [10] has good performance in the image restoration task. Inspired by this, we designed a structure to extract deeper features based on this, three units and an upsampling module are integrated using four residual connections. Then, use bilinear interpolation operators to change the size of the feature so that the output feature has the same dimensions as the original feature. Eventually, these features are connected directly to the front of the spectral attention remaining module (SARM) by skipping the connection. The module SAR adaptively rescales the features of each of their channels. Finally, the reconstruction module composed of four filters will output reconstructed images of the same size as the input. We adopt the ReLU layer after each filter to enhance the representation ability of the reconstruction block. We set the size of all filters to 3×3 (with the stride of 1 and 1 pixel overlap) except for the last filer, where the kernel size is set to 1×1 (with the stride of 1 and nonoverlap). The final convolution layer will compress the features from 128 channels to 102 channels and complete the process of image reconstruction.

This can be expressed simply as follows:

$$H_{\text{Net}}(Z) = f_\theta(\text{IPM}(Z)) \tag{3}$$

where $H_{\text{Net}}(\cdot)$ is the function of the proposed DRPSR approach. $f_\theta(\cdot)$ denotes the network structures with parameters other than IPM, while the upper module is either "nearest" or "bilinear." Other methods can also be used but these prevailed in our experiments. When the random noise is first input to the network, the network has not been trained, so the network parameters are still initial and cannot output meaningful images. In other to measure the difference between the output I_{sr} and x_0, we adopt $l1$ loss. After the back-propagation, the generated I_{sr} will gradually approach the x_0. In this process, the noise will be rejected and then the image with higher quality can be reconstructed. The specific process is shown in Fig. 1. Different from previous DIP-based methods, which are mainly composed of coding–decoding architectures, our network architecture can be more flexible. We divide the network part into the input processing network, main network, and reconstruction network and adjust the coordination between the input and the main network through the input processing part, so as to achieve a more flexible network structure design and better performance of reconstructed images.

B. IPM Architecture

In Fig. 2, we present the network structure of the proposed IPM. It mainly consists of three parts: PA, channel adjustment, and SRA. We denote $Z \in \mathbb{R}^{h \times w \times C}$ the input random noise.

Note that, since we adopt the IPM, different inputs have little influence on the final result, so we still choose random noise for the convenience of comparison. $Z^* \in \mathbb{R}^{h \times w \times C}$ represents the output after IPM processing. Usually, LR HSIs are used as input in the network, and there are certain relationships and rules between the pixels of the images. We use the paradigm of the DIP approach and find the optimal weights of a network by intrinsic prior, which is contained in a network structure instead of the datasets. It means that the network structure in this method is more critical than that in other deep learning-based methods, because it relies solely on the structure to capture prior information to construct images, which also reflects the importance of breaking the DIP algorithm's limitations in structural design. The DIP work is to prove that even noise can be used to reconstruct images, but what is the input that gives the best performance to the network is not discussed in detail, which also limits the application of this method. We find that the lack of reasonable input hinders the design of the network structure of the method and limits the quality of the reconstructed image. Our goal is to obtain input that resonates with the subsequent network structure, so as to improve the SR performance. Based on the subsequent network structure, the IPM will determine what distribution of inputs can be more conducive to improving performance. The optimized input can be obtained via IPM by feeding the random noise Z

$$Z^* = F_{\text{IPM}}(Z) \tag{4}$$

where $F_{\text{IPM}}(\cdot)$ denotes the proposed IPM. Different from previous methods, which directly input random noise, we adjust the input on different levels. We will give details below and present the effect of different parts.

1) Pixel Adjustment: Although the convolution operation with sufficient depth can change the pixel to some extent, it has two disadvantages: First, it is not intended to adjust the value

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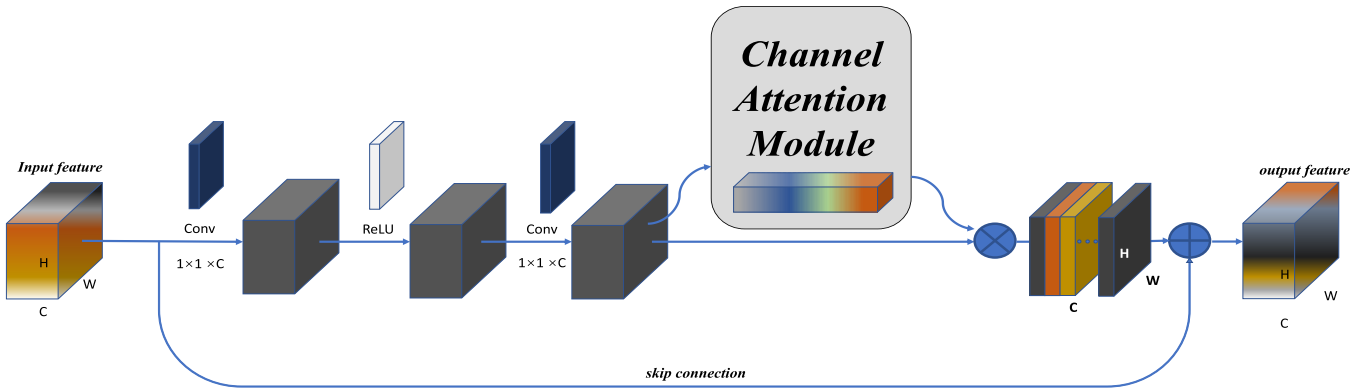


Fig. 3. Network structure of SARM. "+" denotes elementwise addition and "×" denotes elementwise multiplication.

of each pixel, but to adjust the values of the spatial region together with the commonly used convolution kernel (e.g., 3×3), which is not accurate. Second, to achieve good results, you need a very deep network, which will cause a waste of computing resources. In order to make the input image be adjusted by IPM quickly, a PA layer is used to assign weight to each pixel, which can replace many convolution operations and achieve better performance. For this reason, we adopt C filters of size $1 \times 1 \times C$, where a filter operates on one pixel across C channels. Different from channel adjustment, 1×1 filters do not attempt to change the number of channels but simply assign an adaptive weight value to each pixel. This operation is designed to improve the matching degree between the input and the subsequent network and play an important role (as shown in Table IV). Thus, the value of each pixel and the overall distribution are changed in the training

$$Z_p = C_p(Z) \tag{5}$$

where $C_p(\cdot)$ and Z_p denote a PA layer and the pixel-adjusted input, respectively.

2) Channel Adjustment: After PA, we need to stretch and compress the width (the number of channels) as a whole to adjust input. We do it in different branches

$$Z_b^{(1)} = H_{\text{str}_1}(Z_p) \tag{6}$$

where $H_{\text{str}_1}(\cdot)$ refers to a series of convolution operations to stretch width (increase the number of channels). $Z_b^{(1)}$ denotes the input of 128 channels after convolution. Another branch uses $H_{\text{str}_2}(\cdot)$: four filters of the size $1 \times 1 \times C$

$$Z_b^{(2)} = H_{\text{str}_2}(Z_p) \tag{7}$$

we concatenate them together(as shown in Fig. 2), that is, $Z_c = [Z_{\text{up}}, Z_b^{(2)}]$. In this way, we increased the feature width in order to finally compress the number of channels. In this process, spectral attention residual module was used to adaptively assign different weights to different channels to adjust the features from the channel level [please refer to (7)]. It should be noted that the number of channels has reached 132, after SRA [please refer to (9)], so we adaptively rescale the features of each channel by modeling the interdependencies between feature channels

$$Z_{\text{ca}} = H_{\text{SAR}}(Z_s) \tag{8}$$

where $H_{\text{SAR}}(\cdot)$ and Z_{ca} denote the spectral attention residual module (SAR) and the channel-adjusted input, respectively. Z_s is the input after two up-sampling operations. We give the details of the SAR module in Fig. 3. The residual structure and CA structure are further introduced in SARM, the former pass the low-frequency information directly to the end. The CA [36] mechanism translates the global spatial information of a channel into a channel descriptor by using global average pooling. Such channel statistics can be regarded as a collection of local descriptors whose statistics help to express the whole image. The mechanism adaptively adjusts the characteristics of each channel. Moreover, this residual structure makes the residual structure focus on the processing of high-frequency features. In contrast to residual structures, the CA structure plays different roles at the end of the DRPSR and IPM. In IPM, it is mainly to adjust the data structure of channel level to resonate with networks. At the end of DRPSR, we adopt a CA module and several 1×1 filters to form the SARM to take advantage of the correlations between different spectral bands. The CA structure in SARM feeds the processed spatial–spectral features to the next layers, which will be constructed to the SR image. After each convolution layer, a ReLU layer is applied to enhance its representation ability.

3) Spatial Region Adjustment: Further adjust the input by scaling the spatial region

$$Z_{\text{up}} = H_{\text{UP}}\left(Z_b^{(1)}\right) \tag{9}$$

where H_{UP} and Z_{up} denote an upsampling module and up-scaled input, respectively. We leverage the bilinear interpolation operator to establish the upsampling structure. Note that, in the one-step upsampling procedure, it is not easy to construct a high-quality SR image. Therefore, we adopt two or three upsampling modules to handle this process, depending on the size of the image. In addition, we use the BatchNorm [37] to adjust the spatial area data distribution, which plays an important role in the iterative training of random noise input.

We adopt a global skip connection to make the final adjustment of pixels

$$Z^* = F_{\text{IPM}}(Z) = Z_p + H_{\text{cp}}(Z_{\text{ca}}) \tag{10}$$

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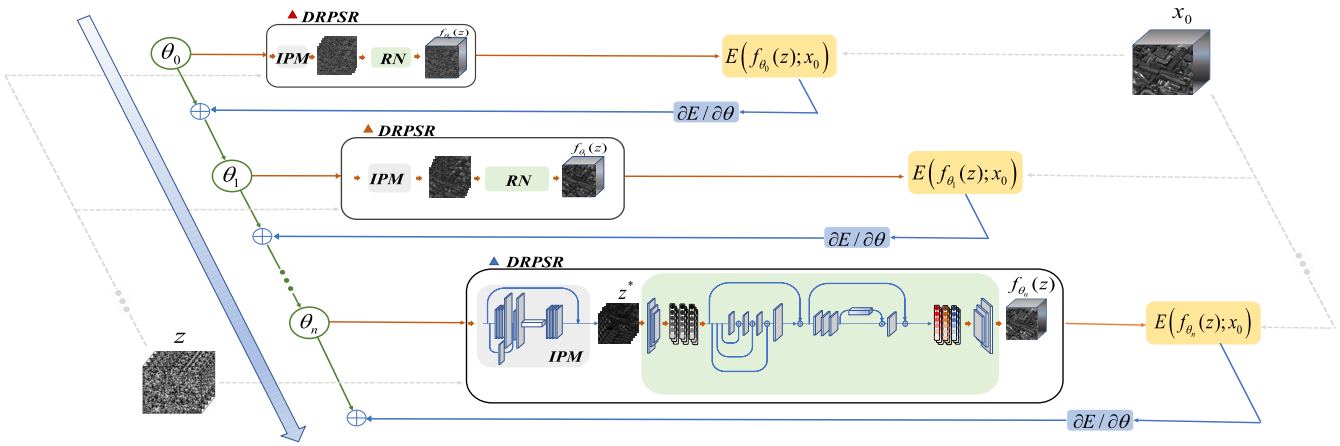


Fig. 4. Visualization of iterative process: image restoration using the DRPSR. We iteratively update DRPSR from an initial weight θ_0 to minimize the (2).

in contrast to H_{UP} [refer to (9)], the spatial region is compressed by H_{cp} (a series of convolution operations). After the elementwise sum operator, the final pixel is added to obtain the optimally adjusted input Z^*. Depending on the size or dimension of the cropped image, the number of the convolution layers can be appropriately increased, to enlarge the receptive field for the image with larger sizes and higher dimensions.

C. Iterative Process

We show the iterative process in Fig. 4. To minimize the data item (2), we continually update a random weight θ_0 in multiple iterations. $x = f_\theta(z)$ denotes an image, where the weights θ are mapped to it at every iteration. z and f are a fixed tensor and a neural network with parameters θ, respectively.

The output Z^* processed by the IPM is continuously optimized with iteration as the input for the next resonant network (RN). The output of the RN is first downsampled and then the error is calculated with x_0 by (13). The error is continuously reduced by back-propagating. A large amount of image statistical prior information is learned through the network structure, whereas noise is rejected.

Noise as an input will hinder the reconstruction of the image, but the interference is different for diverse networks. The experimental results show that although the U-Net-like "hourglass" architecture has a strong ability to resist the interference of noise input, the PSNR of the reconstructed image with DIP + IPM structure is still significantly improved. For many other types of networks, such as DRPSR-IPM (i.e., the subsequent network part), ResNet, and DenseNet, it is difficult to reconstruct the image with good quality by direct input noise. In this way, IPM not only improves the performance of the algorithm, but also learns the characteristics of the algorithm itself, so that it can be applied without limitations.

D. Loss Function

In other to make SR results close to ground-truth images, many cost functions have been investigated, such as l_2, l_1, perceptual, and adversarial losses. l_2 loss or l_1 loss is more credible in the remote-sensing field because some cost functions may introduce details that are not present in the original HSIs. Instead of using l_2 loss in the DIP work, we use l_1 loss in this work, which can maintain better convergence and effectively penalize small errors. Moreover, the final super-resolution performance is better than l_2 in this method.

TABLE I
AVERAGE QUANTITATIVE PERFORMANCE BY DIFFERENT LOSS FUNCTIONS ON PAVIA CENTRE DATASET

Up-scale	Losses	L2	L1
× 4	MPSNR	28.326	28.659
	MSSIM	0.758	0.761
× 8	MPSNR	24.752	24.887
	MSSIM	0.501	0.503

The l_1 loss is defined as follows:

$$\mathcal{L}_1(\Theta) = \frac{1}{N}\sum_{n=1}^{N} \| I_{\mathrm{HR}} - H_{\mathrm{Net}}(I_{\mathrm{LR}}) \|_1 \tag{11}$$

where I_{HR} and $H_{\mathrm{Net}}(Z)$ are the ground-truth HSI and reconstructed HR HSI, respectively.

Our purpose is to reconstruct HSI x from an LR HSI x_0 by our proposed DRPSR. Following (2), we get the data item

$$E(x; x_0) = \| d(x) - x_0 \|_1. \tag{12}$$

Here, d is referred to as a down-sampler, which can resize an image by a factor t. Up to now, some down-sampling methods try to simulate real fuzzy scenes, but due to the complexity of the design, it is difficult to obtain good performance. Besides that, compared with common interpolation down-sampling methods such as bilinear, nearest, or bicubic, complex samplers can easily lead to over-fitting in training. Among the regular interpolation down-sampling methods, we choose the bicubic interpolation method to perform the down-sampling operation to achieve better performance. x can be expressed as $x = f_\theta(\mathrm{IPM}(Z))$. Plugging it in (12)

$$E(x; x_0) = \| d(f_\theta(\mathrm{IPM}(Z))) - x_0 \|_1. \tag{13}$$

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TABLE II
AVERAGE QUANTITATIVE COMPARISONS ON PAVIA CENTRE DATASET BY FACTOR 4 AND 8

		Bicubic	SRCNN [25]	3D-CNN [12]	TLCNN [8]	VDSR [22]	GDRRN [42]	SSPSR[43]	DIP [15]	Ours
× 4	MPSNR↑	27.556	27.643	28.265	27.671	27.867	28.414	29.1581	28.182	28.659
	MSSIM↑	0.694	0.711	0.746	0.708	0.722	0.749	0.790	0.734	0.761
	SAM↓	6.152	6.752	6.002	6.701	6.654	5.574	5.461	6.132	5.983
	RMSE↓	0.044	0.043	0.041	0.043	0.041	0.042	0.059	0.041	0.040
	ERGAS↓	6.895	6.812	6.311	6.908	6.724	6.264	5.801	6.525	6.253
× 8	MPSNR↑	24.595	24.702	24.790	24.532	24.823	24.859	25.199	24.715	24.887
	MSSIM↑	0.472	0.480	0.501	0.478	0.502	0.501	0.537	0.494	0.503
	SAM↓	7.746	7.821	7.782	8.283	8.018	7.366	7.331	7.863	7.618
	RMSE↓	0.062	0.065	0.062	0.063	0.061	0.061	0.019	0.062	0.060
	ERGAS↓	4.830	4.794	4.645	4.932	4.652	4.633	4.312	4.721	4.619

IV. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we show the experimental results to evaluate the effectiveness of DRPSR on two public remote-sensing HSI datasets, that is, the PAVIA CENTRE dataset and CHIKUSEI dataset. We used the ADAM optimizer [38] in the proposed iterative optimization. The proposed method is performed on a server with four GPUs: NVIDIA GeForce GTX2080Ti with 11-GB memory.

A. Remote-Sensing HSI Datasets

Pavia Centre Dataset: It is a public HSI dataset, which is the center area of Pavia, taken by Reflective-Optics-System Imaging Spectrometer sensor in 2001. It has 102 spectral bands and 1096 × 1096 pixels.

Chikusei Dataset: It is the downtown area in Chikusei of Japan, taken on July 29, 2014. It has 2517 × 2335 pixels and 128 spectral bands with the spectral range from 363 to 1018 nm in total. To evaluate the DRPSR approach, we crop the image to obtain a subimage dataset. We choose to use the obtained image as an HR dataset for comparison, which is further degraded as the LR dataset.

Evaluation measurements: The performance of our method was evaluated using five commonly used quantitative image quality indicators (PQIs), including spectral angle mapper (SAM) [39], erreur relative globale adimensionnelle desynthese (ERGAS) [40], root mean squared error (RMSE), structure similarity (SSIM) [41], and peak SNR (PSNR), and we show the mean values of all bands for PSNR and SSIM, referred as MPSNR and MSSIM, respectively. The remaining three indices are extensively adopted in the HS fusion field. The best values for above five metrics are 0, 0, 0, 1, and $+\infty$, respectively.

B. Results on Pavia Centre Dataset and Chikusei Dataset

Tables II and III tabulate the performance of several competing methods in terms of five PQIs, which show that the DRPSR method is superior to DIP algorithms. The performance of this algorithm is leading in the nontraining-based methods and comparable to many classical training-based methods. To demonstrate the performance of the DIP algorithm, we select the network that Sidorov *et al.* [15] applied the DIP algorithm to the HSI SR task, which is more suitable for HSIs in the processing of channel numbers. Compared with DIP, our method surpasses it in all the objective evaluation indexes and achieves the goal of improving DIP work. Compared with the method requiring training (VDSR [22], TLCNN [8], 3-D-CNN [12], and SRCNN [25]), our method also surpasses them in all the indexes of reconstructed images and achieves the purpose of applying it to HSI super-resolution task. In comparison with the GDRRN [42], we are only behind on the SAM metric. We also show some visualization results of the compared methods on Pavia Centre and Chikusei datasets, as illustrated in Figs. 5 and 6. On the one hand, this is due to the superiority of GDRRN itself in this metric; on the other hand, the method of image recovery in random noise has its limitations in some parts. The algorithm is still difficult to compete with recently proposed train-based approaches such as SSPSR. We propose the DRPSR method to solve the difficulties of unclear reconstructed images and low indicators caused by insufficient training data in small datasets and improve the application of the DIP method. The advantage of DRPSR lies in that it does not need to be trained based on samples and ignores the differences in imaging conditions between different datasets. This work does not need ground truth and focuses on the scenarios in that it is difficult to obtain training data, which is practical in application. Besides that, we also exploit the application effect of IPM in the natural image method. Although DRPSR is designed for SR reconstruction of HSIs, compared with DIP on natural images, this method still has a better performance. Through experiment, we found that IPM can still play an important role in making the network structure unrestricted and improving performance in the gray/RGB image SR field.

C. Ablation Studies

The improved performance of the DRPSR method is mainly due to the optimization of the input by IPM and the special design of the subsequent network structure, to take advantage of the spatial characteristics and spectral correlation information of the HSIs. Here, we focus on IPM that contains three main components including PA, channel adjustment, and SRA. We modify the structure of DRPSR and compare the variants to verify the effectiveness of each component.

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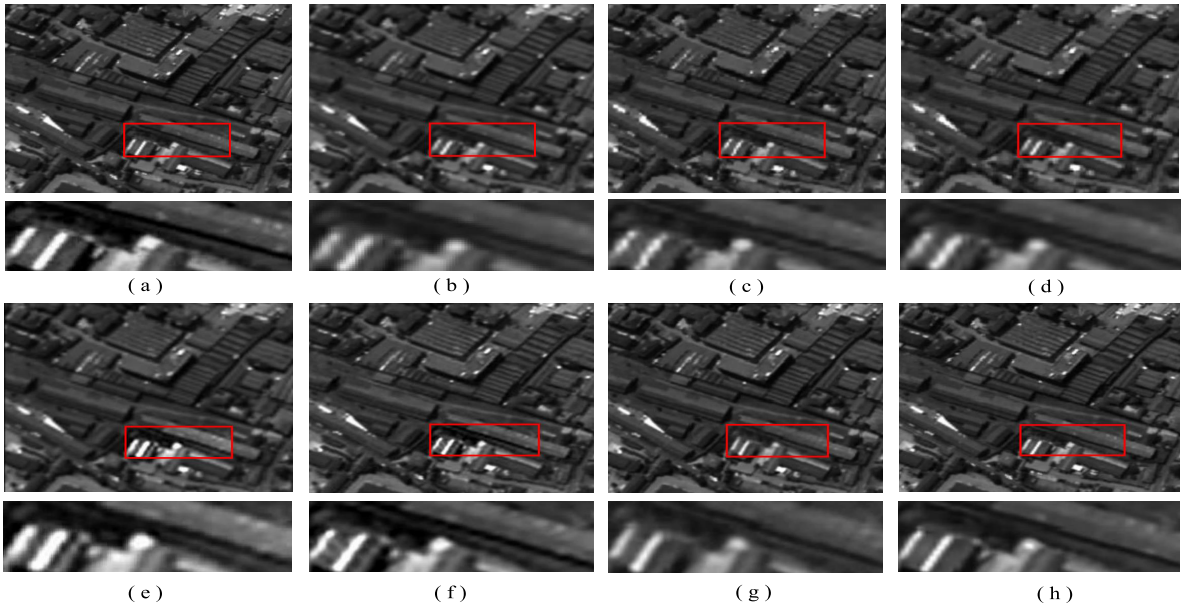


Fig. 5. HSI super-resolution results of different methods on PAVIA CENTRE image; rescale factor 4; spectral band 25 visualization. In order, they are (a) ground truth, (b) Bicubic, (c) TLCNN [8], (d) VDSR [22], (e) 3-D-CNN [12], (f) GDRRN [42], (g) DIP [15], and (h) proposed DRPSR.

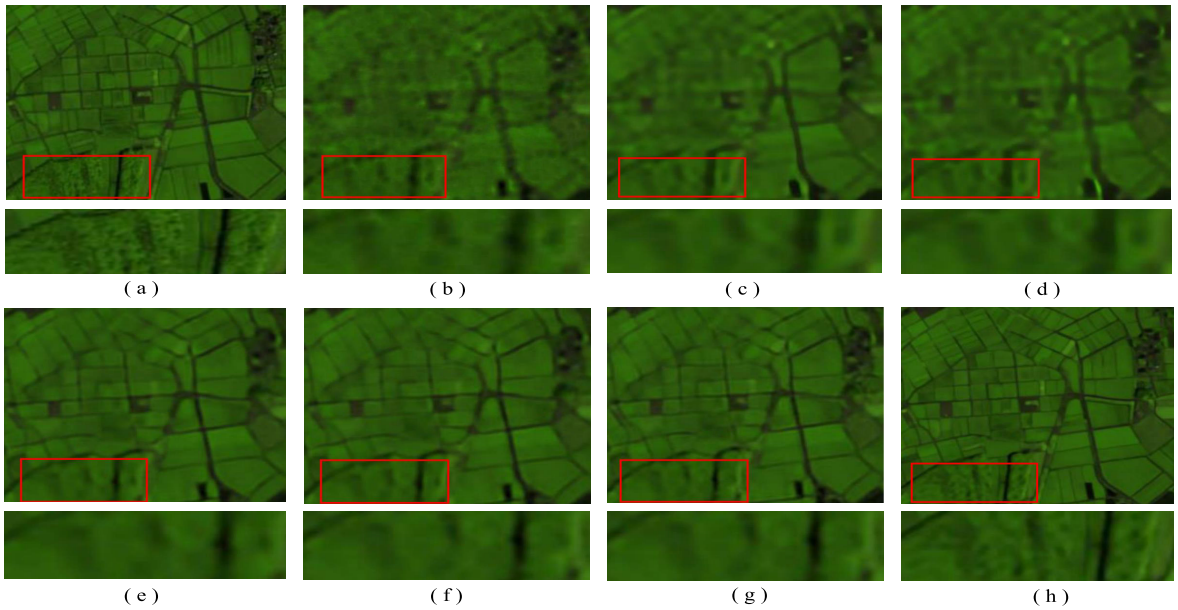


Fig. 6. HSI super-resolution results of different methods on CHIKUSEI image; rescale factor 4; false-color with spectral bands (65, 100, 35). In order, they are (a) ground truth, (b) Bicubic, (c) TLCNN [8], (d) VDSR [22], (e) 3-D-CNN [12], (f) GDRRN [42], (g) DIP [15], and (h) proposed DRPSR.

We use the image from the PAVIA CENTRE dataset as ground truth and evaluate the SR performance.

In Table IV, we present the performance of IPM under ablation experiments. In this part, the proposed DRPSR network structure is fully applied to the PAVIA CENTRE dataset, and we adjust different factors to carry out the experiment.

The case of "IPM" denotes the performance of the DRPSR method without deleting any part of the IPM. As you can see from the data, a full IPM delivers the best performance in the table. It can be proved that the input processed by the IPM has indeed played a "resonance" effect with the following network structure. In the same way, "IPM-w/o PA" means the IPM removes the PA module. As there is only one layer of structural difference, the effect is relatively small compared with other experiments. But compared to the first experiment, there is no denying that its role is still huge. In channel adjustment structure, which is composed of many parts, we mainly remove the CA module structure in Fig. 3, that is, the case of "IPM-w/o CA" and it can be seen that the performance is much lower than the second experiment. The example with the most degradation is "IPM-w/o SRA," we remove the progressive upsampling scheme and batch normalization. Compared to the one-step upsampling scheme, we leverage the progressive upsampling SR framework inspired

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TABLE III
AVERAGE QUANTITATIVE COMPARISONS ON CHIKUSEI DATASET BY FACTORS 4 AND 8

		Bicubic	SRCNN [25]	3D-CNN [12]	TLCNN [8]	VDSR [22]	GDRRN [42]	SSPSR[43]	DIP [15]	Ours
× 4	MPSNR↑	37.528	37.643	38.297	37.698	37.763	38.584	40.361	38.216	39.302
	MSSIM↑	0.889	0.896	0.911	0.901	0.904	0.913	0.941	0.901	0.926
	SAM↓	3.524	3.811	3.286	3.782	3.614	2.674	2.353	3.411	3.112
	RMSE↓	0.016	0.016	0.015	0.016	0.015	0.011	0.011	0.015	0.013
	ERGAS↓	6.756	6.747	6.101	6.752	6.891	6.089	4.989	6.153	6.007
× 8	MPSNR↑	34.495	34.406	34.807	34.361	34.537	34.820	35.837	34.711	34.908
	MSSIM↑	0.821	0.818	0.829	0.816	0.831	0.835	0.862	0.829	0.841
	SAM↓	5.028	5.289	5.039	5.318	5.192	4.357	4.013	5.339	5.013
	RMSE↓	0.023	0.024	0.022	0.022	0.023	0.021	0.019	0.021	0.019
	ERGAS↓	4.851	4.897	4.651	4.885	4.912	4.593	4.312	4.681	4.583

TABLE IV
ABLATION STUDY ON PAVIA CENTRE DATASET. PA, CA, AND SRA REPRESENT PA, CHANNEL ADJUSTMENT, AND SRA, RESPECTIVELY

Up-scale	Models	MPSNR↑	MSSIM↑	SAM↓
× 4	IPM	**28.659**	**0.761**	**5.983**
	IPM-w/o PA	28.243	0.760	6.388
	IPM-w/o CA	27.996	0.758	6.671
	IPM-w/o SRA	27.568	0.755	6.932
× 8	IPM	**24.887**	**0.503**	**7.618**
	IPM-w/o PA	24.792	0.481	7.873
	IPM-w/o CA	24.616	0.479	7.991
	IPM-w/o SRA	24.473	0.475	8.165

by [23], which decomposes a complex task into some simple tasks to reduce the learning difficulty. The latter BN operation has been proved to be effective in many works, and it is irreplaceable in this work with irregular data structure as input.

It is worth noting that if we remove the IPM structure, but directly use random noise as input, then the performance of subsequent RNs will be very poor or even unable to produce meaningful images. Therefore, we did not carry out the comparison experiment of removing IPM. This does not mean that the RN (i.e., DRPSR-IPM) part of the DRPSR network is not as good as the DIP, but this part of the structure is not able to resonate with the random noise input. It is difficult to reconstruct a clear image if the input image does not match the network structure. This is why the SR performance is often different when the input image is different, especially for the network structure with residual modules. IPM focuses on the resonance effect between input data structure and subsequent network structure. When different images are input into the DRPSR network structure, the SR performance changes to a more stable value compared to the DIP method. We compare the performance of the DIP-based network under the same scaling factor in Tables II, III, and V, IPM brings considerable performance improvements to the DIP, for example, $+0.477$ dB for $\times 4$ and $+0.172$ dB for $\times 8$. This proves that the introduction of IPM not only increases the choice of network structure, but also is more conducive to the representation of spectral features. We show the visualization results of IPM output under different iterations in Fig. 9. At the beginning of the iteration, the output image of the IPM module changes significantly. As the number of iterations gradually increases, the changes in the output image are almost invisible to the naked eye. It can be seen from the figure that the optimized input graph generated after 1000 iterations has similar characteristics to the real image. At this time, the optimized feature map can be directly used as the input image and input into the main network without IPM to obtain a high-quality reconstructed image as Table V. HSIs often contain multiple feature information in a single-pixel point, and it is difficult to distinguish the quality of false-color images by human eye vision. Therefore, we show mean absolute error maps for the reconstruction results of each algorithm in Fig. 7, and the error values represented by each color are shown on the right, with the closer the image is to blue indicating the higher spectral quality of the reconstructed image. The results of the DRPSR proposed in this article are significantly better than the reconstructed image of the DIP algorithm, with good performance even in complex urban building and road areas, achieving results that approximate those of the GDRRN algorithm that requires training.

D. Global Iterative Trend and Stability of Performance Analysis

To facilitate the presentation of DRPSR and DIP performance during the iteration, we adopt a more intuitive method, as shown in Fig. 8. In the two figures above, we employ mean values of all spectral bands of two PQIs to evaluate the performance, including PSNR and SSIM [41]. After the image quality of network reconstruction reaches the peak, we need 40 000 iteration time to carry out the experiment, so as to ensure that the results produced in the experiment process were approximately the best, and at the same time, it was convenient for us to observe the stability of the algorithm in the iteration. We record experimental results after every 500 iterations and report the average precision among ten experiment results. After about 2500 iterations, the network reconstructed relatively clear images and then started comparative observation: we used DRPSR (blue line), DIP (red line), and DIP + IPM (black line) to do a comparison test.

As we can see, in the end, after 40 000 iterations, the difference between the peak and the lowest PSNR of our method

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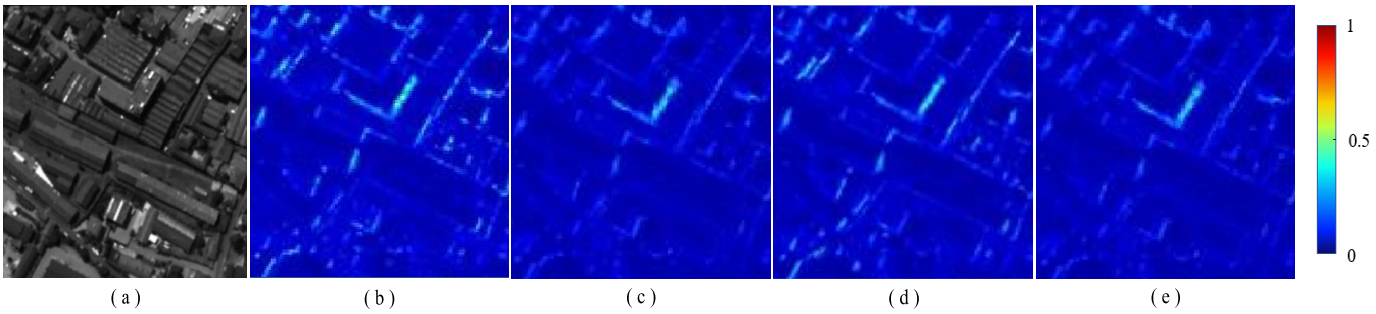


Fig. 7. Corresponding spectral angle mappings of different upsampling methods. (a) Reference. (b) TLCNN. (c) GDRRN. (d) DIP. (e) Proposed DRPSR.

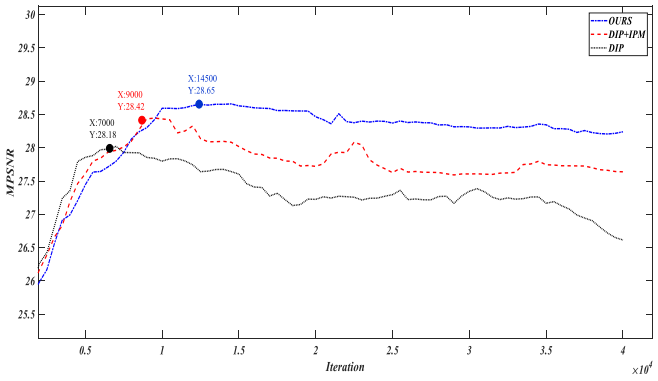


Fig. 8. Global iterative trend and stability of performance comparison; On PAVIA CENTRE DATASET by factor 4.

is much smaller than the difference between the highest and lowest PSNR in DIP. Although the blue line reaches the peak with more iterations than DIP and DIP + IPM, this is due to the addition of the IPM and our more complex network architecture, which means we have more network parameters than DIP. The best performance of DIP is achieved when iteration equals 7000 as shown in Fig. 8, then DIP + IPM achieves the best performance at iteration = 9000, and IPM improves the performance of DIP reconstruction images. Finally, it can be observed that DRPSR reached the best PSNR at iteration = 14 500. Although it takes more time to reach the best performance than the DIP approach, it is noted that it greatly improved the reconstructed image eventually, and also the DRPSR has already surpassed the image quality reconstructed by the DIP network after about iteration = 8000. In practical application, it may be difficult to determine the best stopping point because there is no ground-truth image. Some general unreferenced image quality evaluation methods based on transform domain and space domain can be used to find the generated high-quality image, but there is still a simpler method in the application. Since the DRPSR network confirms the range of optimal stopping points in the experiment and the IPM makes the results less prone to degradation after several iterations, the reconstructed images selected after this can achieve good results. Based on the above-mentioned analysis, it is obvious that the network with IPM will have a better effect on the final reconstructed images, which can also prove the effectiveness of IPM.

Ulyanov found naturally looking images resulted in much faster convergence, and the noise was rejected by means of comparative experiments. After the noise is output by the IPM, it has a relatively natural texture that is matched with the subsequent resonated network, which greatly improves the texture clarity of the reconstructed image and better eliminates the influence of noise. The function of IPM is not limited by improving the stability of network iteration and preventing it from degrading, after the network parameters become stable after several iterations. Finally, IPM will output an optimized input - Z^* during the iteration process, which can be considered as the deep features of original HSIs.

Instead of interpolated low-resolution images or some other type of input, we chose Z^* as the input. We tried to directly input the noises or the LR image, but in most cases, they failed to be reconstructed to be a natural-looking image. In this approach, it does not matter whether the data are structured in a regular way, rather that they fit into the network structure. LR image as an input does not introduce more prior information, thereby giving I_{sr} gain. Generally speaking, due to the complex structure of data, it is difficult to obtain satisfactory reconstructed images by directly inputting random noise into the network without IPM. This is due to the increasing number of network parameters used to process HSIs and the addition of many residual structures makes the structure of the data hard to resonate with the network structure. We finally obtain the higher quality image by adding IPM, which reflected the importance of adapting network parameters to the data structure. Although the work of DIP shows that the structure of generator ConvNets itself has the ability to capture a large number of image statistics. A deep network theoretically eventually can be adaptive to adjust the input to the reconstruction of the high-resolution image in this method, but be aware that if we do not have the firm purpose to adjust the structure of the data, in most cases, we end up with a network that is too deep and contains too many parameters to iterate over. Without input processing, it is difficult to find a network structure that can match well with the nonstructured noise, and the reconstructed image is not satisfactory, this will greatly limit the application of the DIP algorithm. We did the following comparison experiment (as shown in Table V) to illustrate the reasons more intuitively.

Because the type of input has a big impact on the choice of subsequent network architecture, the selection of network structure has an important effect on how the method searches the solution space. If the input selection does not match the subsequent network, the performance will be extremely

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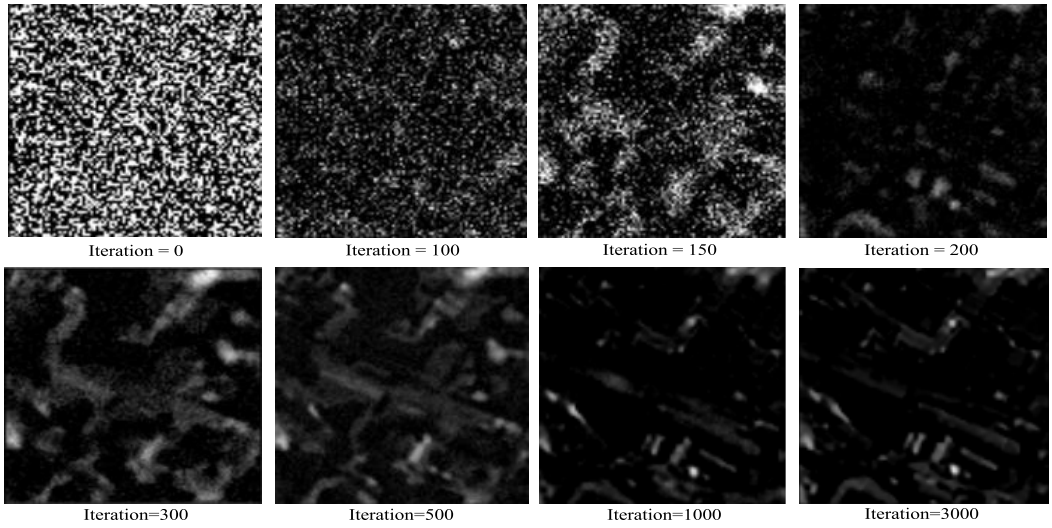


Fig. 9. Visualization of IPM output under different iterations.

TABLE V
QUANTITATIVE COMPARISON RESULTS OF DIFFERENT

Up-scale	INPUT TYPE	MPSNR↑	MSSIM↑	SAM↓
× 4	random noise	28.182	0.734	6.318
	LR image	27.593	0.698	6.443
	random noise/LR image	27.744	0.701	6.391
	stable Z^*	28.411	0.751	6.140
	IPM Z^*	**28.416**	**0.752**	**6.079**
× 8	random noise	24.715	0.494	7.937
	LR image	24.593	0.476	8.076
	random noise/LR image	24.675	0.483	8.113
	stable Z^*	24.811	0.498	7.928
	IPM Z^*	**24.823**	**0.498**	**7.883**

poor. To facilitate the comparison with the original DIP work, we use the DIP hourglass network as the basic network. Finally, by leveraging the IPM, we proposed to show the effect of this module. We use the same random noise as the DIP work, the LR image (I_{LR}) after image interpolation, the superposition image of the above two (we replace part of the channels in the LR image with random noise), the image Z^* output after the IPM parameter is stabilized in DIP network (stable Z^*), and the output of the DIP network with the input optimization module is added (IPM Z^*) as the comparison respectively (Table V).

Throughout the experiment, we find that the reconstructed image using the direct input of the LR image will introduce more low-frequency information, which is closer to the low-resolution image itself, far from our requirements. In order to remove the phenomenon that the reconstructed image is more similar to the LR image, we try to introduce noise to replace the part of the channels in the LR image to destroy the high similarity between the reconstructed image and the input LR image (random noise/LR image), but the final performance is still not ideal. On the contrary, introducing high-frequency information into the input can have a positive effect on the reconstructed image. After adjusting the input with IPM, the performance of the network is greatly improved. Compared with the experiment of using fixed input Z, we speculate that the Z found by IPM contains similar information to the final reconstructed image, and even if the fixed Z is input alone, the image of higher quality can be reconstructed. This indicates that making the input resonate with the subsequent network structure is an efficient way to improve performance. Unlike many existing deep prior-based algorithms that mainly use the U-Net structure, this experiment uses IPM to process the input to the network early in the network, thus breaking its limitations on the network structure. By designing different inputs, we demonstrate that the optimized Z^* can be exploited to infer the missing high-frequency components of original images.

V. CONCLUSION

In this article, we propose a powerful deep resonant prior knowledge-based approach to compensate the DIP algorithm for the lack of IPM. By introducing "IPM" to adjust the input image, the selection of the subsequent network structure is no longer limited. Moreover, the optimized input contains high-frequency information, which further improves the performance and stability. In addition, we develop a stable and effective network architecture DRPSR for HSI recovery. Our method can also dramatically improve performance in the gray/RGB image super-resolution reconstruction task. In most practical applications, many SR methods are almost impossible to apply under complex conditions, and the method proposed in this article can be better applied to this situation. For scenarios where training data and ground-truth references are difficult to obtain, the DRPSR algorithm can efficiently obtain results comparable to those of training-based methods, which is more practical in application. Sufficient experimental results on open datasets verify that our proposed IPM can greatly expand the application of this nonsample-based training method. Furthermore, our proposed DRPSR method has a promising application in the field of HSISR.

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